Page of Contents

Sl.Order	Particulars
A	**Purpose**
B	**Scope**
C	**Responsibility**
D	**Policy :** 1. **Staff and Patient Education and Training** 2. **Guideline Instruction** A. **Safe Work Practices.** B. **Decontamination** C. **Waste Disposal** D. **Laboratory Area.** E. **Laboratory Accidents**

©Hope Hospital

A. **Purpose**: To establish guidelines for good laboratory practices with standard precautions so as to reduce the risk of occupational hazards and enhance safety of staff and patients.

B. **Scope:** Hospital wide.

C. Responsibility: Department of Laboratory Medicine

D. Policy:

1. Staff and Patient Education and Training:

a. All new employees (fresh recruit/transferred) are trained about the safety guidelines observed in the department during their departmental orientation.

b. Periodic Training of the laboratory staff by internal and external trainers are organized relating to better work practices , proper handling of equipment etc.

c. Patients are educated regarding the laboratory tests to be undertaken by them , special precautions or instructions if any to be observed prior to and during the test etc.

2. Guideline Instructions:

A. SAFE WORK PRACTICES

1. **HANDLING OF SPECIMEN :**

a. Gloves

- Wear gloves and laboratory coats (aprons) at all times when handling and processing patient specimen, decontaminating instruments and cleaning.
- Bandage open cuts and scratches on the hand and then wear gloves.
- Wear gloves when performing phlebotomy and handling actual blood specimens.
- Wash hands immediately after gloves are removed , after a task that involves heavily contaminated matter and before leaving the laboratory

b. Specimen Transport

- Always transport specimens to the laboratory in leak proof containers to avoid hemolysis.
- Do not accept grossly soiled or contaminated specimens. Notify the individual responsible for submitting such a specimen and follow the laboratory specimen rejection policy.

c. Needles and syringes

- Use needle-locking syringes or plastic disposable syringe-needle units.
- Never bend the needles, after use do not recap and cut the needle with help of needle destroyer, discard them in the *sharps container*.
- Secure blood culture bottles before inserting needles into the bottle (eg., place bottle in support rack).

d.Tubes

- Always carry tubes in racks
- Use plastic tubes when possible
- Uncaps tubes carefully, avoid splashes or sprays (eg. when removing tops from vacuum tubes). In case of splash or spray, see *the shower and eyewash* section in this document.
- Do not use glass tubes that are broken or damaged at the mouth. Discard such tubes into the sharps container.

e. Centrifuges

- Centrifuge tubes must be intact and properly balanced when centrifuged
- Do not place tabletop centrifuges in the biological safety cabinets
- Clean the centrifuge once daily after use to remove any contaminating material on the inner side of the centrifuge.

f. Handwashing

- Frequent hand washing after removing gloves, before leaving the laboratory are absolutely essential

- Use non irritating soap for routine washing
- Use antiseptic soap or an alcohol based hand disinfectant followed by thorough hand washing for accidental skin contamination.

2. **HANDLING CHEMICALS**

- Wear appropriate PPE when handling hazardous chemicals.
- Label all reagents with their chemical names and appropriate hazard warnings provided from their material safety data sheets (MSDS).
- Keep MSDS for all chemicals either in the laboratory or in the office near by.
- Store all hazardous chemicals, including chemicals, reagents and dyes, below eye level.

3. **HOUSEKEEPING AND MISCELLANEOUS SAFE PRACTICES**

General

- Avoid or minimize activities associated with transmission of infectious agents
- Designate clean and contaminated work area
- Clean and disinfect (see section on decontamination) all surfaces after spills and at the end of each work shift
- Keep all work areas neat and uncluttered
- Do not store personal items in the work area
- Do not bring food or beverages into the laboratory area
- Remove coats before leaving the laboratory
- Dispose all contaminated materials in appropriate coloured closed bags

B.<u>DECONTAMINATION</u>

Routine decontamination and cleaning of the work environment are the responsibility of all laboratory workers particularly, of the housekeeping staff. To accomplish this work area should be uncluttered, with clean and unclean materials clearly demarcated and separated. The section below outlines the common decontamination protocols to be followed in the routine day-to-day functioning of the laboratory.

1. PROCEDURES FOR DECONTAMINATION

a. Preparation Of 10% Household Bleach

- Prepare fresh daily
- Add one part of household bleach to nine parts of tap water
- Dispense in wide mouthed large plastic containers. The containers should be only half full
- Place the containers in the designated work areas

b. Decontamination of Work Surfaces

- Works surfaces have to be decontaminated at least twice daily, before the work begins and at completion of work
- Use a paper towel or a soft cloth soaked with the disinfectant (1% sodium hypochlorite solution)
- Wipe the work surface going over each area at least twice. Allow to air dry with a minimum contact time of 5-10 min
- The housekeeping staff have to regularly fill the decontamination worksheet

c. Decontamination of Equipments (including vortex, centrifuge, and Telephone and Computer keyboards)

- Follow same procedure as for work surfaces after consulting the supervisor that the disinfectant is compatible with the equipment surface
- The periodicity of decontamination for the above mentioned equipment has to be decided by the supervisors in each section of the laboratory based on the use of equipment

  NOTE: Do not use alcohol on equipments that are close to open flames (burner); Do not use sodium hypochlorite on metal parts because they may cause rusting; Do not use aldehyde based disinfectants as general surface disinfectants. Use Lysol solution for steel and other metal items.

d. Decontamination of Spills

i. Major spills (with possible aerosol formation)

- Evacuate the area or room and alert all personnel regarding the spill and take care not to breathe in aerosolized material
- Close doors to the affected area and keep it closed for 30 min
- Only the designated staff have to enter the area to clear the spill and the staff cleaning the spill should ensure that they use the appropriate PPE (gloves, mask)
- Use disposable paper towels or tissue to wipe off the liquid in the spillage and discard the tissue into a container meant for infected wastes
- Carefully clean the spill site of any visible material from the edges of the spill to the center with a aqueous detergent solution
- Pour disinfectant (1% sodium hypochlorite) over the entire area of the spillage and let it remain for 20 min

- Absorb the detergent with an absorbable material and dispose in the infected container
- Rinse the spill site with soap and water and air dry
- Discard the gloves and mask used for clearing the spillage site into the container for infected items
- Wash hands with soap and water

ii. Minor spills:

- Similar to the procedure for major spills except evacuation of personnel working in the area may not be essential

C. WASTE DISPOSAL

Proper waste disposal and management of biohazardous wastes is essential to limit the risk of infections to the health care workers. Hospital wastes include infectious and non infectious items that have to be disposed using the following steps:

1. Segregation
2. Transport
3. Temporary storage
4. Final disposal

While the laboratory may not be directly involved with steps 3 and 4, segregation of wastes at source and proper/timely transport of the wastes to the terminal area will remain a responsibility of the laboratory

1. Segregation:

- Segregation of wastes generated in the laboratory is the responsibility of every laboratory worker
- The housekeeping staff are responsible for ensuring that every section of the laboratory has disposal containers with all the bags as mentioned below

- The following colour code has to be strictly followed while the wastes are segregated
    a. **RED**- INFECTIOUS MATERIAL
    b. **BLUE**- PLASTIC WASTES AND SHARPS
    c. **BLACK**- GENERAL WASTES
    d. **YELLOW**- ANATOMICAL PARTS

- Wastes are to be segregated as follows;

    a. Wrappers of gloves, paper, wrappers of syringes etc are considered general wastes and are to be disposed in the **black bags.**

    b. Empty saline bottles, empty media/reagent plastic bottles/vials, plastic test tubes that do not contain any infectious material etc. All needles, broken vials, and broken slides that are not to be reused are referred to as plastic wastes and are disposed in the **blue bags.**

    c. Inoculated Petri dishes, clinical samples/specimens including blood and swabs containing infected material, used gloves are considered infectious and have to be disposed into the **red bag**. Any tissue or paper towels used

to decontaminate surfaces in case of spills are also disposed into this container. Dressings, cotton and gauze pieces if used in the OP collection area are also disposed in a similar manner.

d. Anatomical waste, tissues and other body parts are to be disposed in yellow **bag.**

DO NOT DISPOSE ANY SHARP OR BROKEN GLASSWARE INTO ANY OF THE BAGS FOR DISPOSAL.

NOTE: NOTIFY THE HOUSEKEEPING STAFF TO EMPTY SHARPS CONTAINER WHEN IT IS HALF FULL.

The housekeeping staff has to ensure that the bags are emptied and fresh bags are replaced when any of the disposal containers are more than half full. **Wastes are to be emptied from the lab at least twice daily.**

2. Transport:

- All bags that are being transported to the central waste receiving terminal will have to be tied at the mouth to avoid spillage during transport
- Bags should be picked up by the neck and then transported
- Avoid the transport of too many bags at one time and contact of the bag with the body
- Avoid mixing of segregated wastes

D.LABORATORY AREAS

The laboratory should have designated areas for laboratory work and access to such areas should be limited to the laboratory employees.

1. Visitors in laboratory areas

- Children under 12 years of age should not be permitted into any laboratory
- Each laboratory supervisor is responsible for the safety of adult visitors to his or her laboratory.

2. Hazard warning signs and labels

The following four symbols may be used by the laboratory to communicate the extent of risk in the laboratory

1. **NOTICE-**states a policy related to safety of personnel or property but not a physical hazard

2. **CAUTION-**indicates a potentially hazardous situation, which may result in minor or moderate injury

3. **WARNING-** indicates a potentially hazardous situation, which may result in death or serious injury
4. **DANGER-**indicates an imminently hazardous situation that if not avoided, will result in death or serious injury

E. LABORATORY ACCIDENTS

Though the laboratory provides all facilities to avoid accidents, it is inevitable that a few accidents may occur over a period of time. The following section provides details of the management of such laboratory accidents:

1. **Needle stick injuries (including other sharps):**

   - Wash the area immediately with soap and water
   - Briefly induce bleeding from the wound
   - Wash the area again with soap and water
   - Inform the supervisor of the incident and an incident report will have to be made.
   - The supervisor has to assess the risk to the individual and inform the lab director of the accident
   - The hospitals' policy of Post exposure prophylaxis will have to be followed and the appropriate prophylaxis instituted as early as possible

2. **Eye wash station:**
   - There should be at least one eyewash facility per laboratory
   - Laboratory workers exposed to dangerous chemicals or infectious materials should immediately use the eyewash to flush out the material
   - Hold the eyelids open while using the eyewash and let the water spray of for an extended period of time (15 minutes)
   - Report the incident to the supervisor

3. **Incident report:**

   - While it may not be possible to alert the supervisor about the laboratory

accident immediately, the laboratory worker should ensure that the Supervisor is informed about the incident at the earliest.

- The supervisor is duly advised to prepare a report of the incident, using the incident report worksheet and the laboratory director has to be informed .

- The laboratory Incharge will have to ensure that the appropriate PEP/treatment has been followed

5. Fire safety:

Small bench-top are more common in labs than large fires. Labs, especially those using solvents in large quantities have a high potential for flash fires, explosion, rapid spread of fire and high toxicity of products of combustion

a. Prevention

- Plan work. Majority of lab fires have resulted from mental or procedural errors or carelessness
- Keep work area uncluttered and clean
- Observe proper safety practices.
- Store solvents properly. Keep flammable reagents only in explosion proof cabinets
- Wear proper clothing and personal protective equipment
- Avoid working alone, whenever possible
- Read the written emergency plan and be prepared for lab fires even before the event

- Attend the training organized by the lab for use of emergency equipment (fire extinguisher) when needed.

b. Emergency

KNOW WHAT TO DO- go by the emergency plan of the lab planned during the training sessions

KNOW WHERE THINGS ARE- be aware of the closest fire extinguisher, hospital emergency services/ maintenance department telephone number, emergency shower, first aid kit and exit(s)

c. Fire Procedure

- **Notify-** other occupants of the immediate space (yell)
- **Evacuate-** the immediate area of the fire if necessary
- **If safe to do so-** attempt to extinguish

d. Fire Extinguishers

(NOTE: an extinguisher is a first aid tool- do not expect it to control a big fire**)**

To use a portable extinguisher remember the acronym P.A.S.S.

- **P**ull the pin
- Aim extinguisher nozzle at the base of the flames
- **S**queeze trigger while holding the extinguisher upright
- **S**weep the extinguisher from side to side, covering the area of the fire.

Classification of fire and fire extinguisher requirements.

Type of hazard	Class of fire	Recommended extinguisher agents
Ordinary Combustibles Wood, Cloth, Paper	A	Water, dry chemical foam, loaded steam.
Flammable Liquids and gases Solvents & greases, natural or manufactured gases.	B	Dry chemical, carbon dioxide, loaded system, halon 1211 or 1301 foam
Combinations of hazards Ordinary combustibles and flammable liquids and gases.	A & B	Dry chemical, loaded steam, foam.
Ordinary combustibles and electrical equipment.	A & C	Dry chemical
Flammable liquids and gases and electrical equipment	B & C	Dry chemical, Carbon dioxide, halon 1211 or 1301 foam
Ordinary combustibles and flammable liquids and gases, & electrical equipment,	A,B & C	Triplex dry chemical.